



NORTH SYDNEY BOYS HIGH SCHOOL

EXTENSION 1 MATHEMATICS

2015 Preliminary Course Assessment Task 3 (Yearly)

Friday August 28, 2015

General Instructions

- Working time – 2 hours (plus 5 minutes reading time).
- Use a blue or black pen. Diagrams may be sketched in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt all questions.

Section I

- Mark answers on the answer grid provided.

Section II

- Commence each question on a new page.
- Write on both sides of the paper.
- Show all necessary working for every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Teacher (please ✓) ☐ Mr Hwang ☐ Mr Weiss ☐ Mr Ireland
☐ Mr Lin ☐ Ms Ziazaris ☐ Ms Lee

Question	1-5	6	7	8	9	10	11	12	Total	%
Marks	$\overline{5}$	$\overline{8}$	$\overline{12}$	$\overline{9}$	$\overline{13}$	$\overline{12}$	$\overline{12}$	$\overline{9}$	$\overline{80}$	$\overline{100}$
Working out grade:					Legibility grade:					

Section I

5 Marks

Attempt Questions 1 to 5

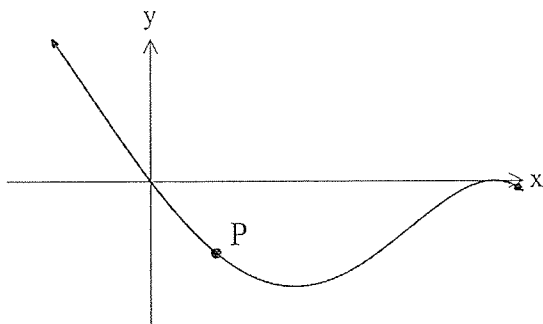
Mark your answers on the answer sheet provided on page 2.

1. Which of the following is a factor of $8x^3 + 27$?
 - A) $x - 3$
 - B) $x + 3$
 - C) $2x - 3$
 - D) $2x + 3$

2. Let α and β be the roots of the equation $3x^2 - 7x + 12 = 0$. What is the value of $\alpha + \beta$?
 - A) $-\frac{7}{3}$
 - B) $\frac{7}{3}$
 - C) 4
 - D) 7

3. What is the value of :
$$\lim_{x \rightarrow \infty} \frac{x^2 - 6}{x^2 - 3}$$
 - A) -6
 - B) 1
 - C) 2
 - D) No Limit

4. Which of the following is equivalent to $\cos(A - B)$
- (A) $\cos A \cos B (\cos A - \cos B)$
 - (B) $\cos A \cos B (\cos A + \cos B)$
 - (C) $\cos A \cos B (1 - \tan A \tan B)$
 - (D) $\cos A \cos B (1 + \tan A \tan B)$
5. Which of the following is true for the point P:



- (A) $f'(x) > 0, f''(x) > 0$
- (B) $f'(x) > 0, f''(x) < 0$
- (C) $f'(x) < 0, f''(x) > 0$
- (D) $f'(x) < 0, f''(x) < 0$

END OF SECTION I

Section II

Total of 75 marks

Attempt Questions 6 to 12.

Write your answers in the writing book provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (8 marks)

- (a) Sketch the following graphs on the **same set of axes**: You should allow $\frac{1}{2}$ a page for your diagram.

(i) $y = 3^{x-1}$ 2

(ii) $y = 4 - 3^{x-1}$ 2

- (b) Given the function 2

$$f(x) = \frac{3x^2 - 2}{x^3},$$

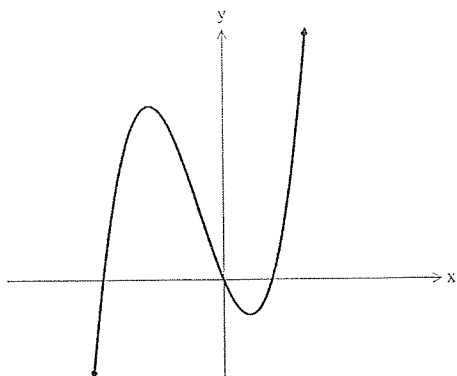
is $f(x)$ an odd function, an even function or neither? Show your working.

- (c) Sketch the graph $y = |x^2 - 16|$ 2

Question 7 (12 marks)

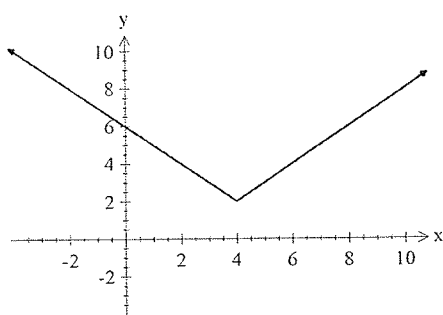
- (a) Copy the following graphs into your workbook, then sketch the gradient function on a set of axes directly below each graph:

(i)



2

(ii)



2

- (b) Differentiate $y = 3x^2 - 2x$ from first principles.

3

- (c) Differentiate and fully simplify the following:

(i) $y = 3\sqrt{x}$

1

(ii) $y = (4 - 3x)^5 \cdot (2x + 1)$

2

(iii) $y = \frac{x^2 + 2}{x + 4}$

2

Question 8 (9 marks)

- (a) Find the values of k for which $x^2 - kx + 4 = 0$ has:

- (i) One root equal to -1 .

1

- (ii) Real roots

2

- (iii) One root which is double the other root.

2

(b) Consider the parabola $-12y = x^2 - 6x - 3$

(i) Find the equation of its directrix.

2

(ii) Find the coordinates of its focus

2

Question 9 (13 marks)

(a) Solve $|2x - 3| = 3x + 7$

2

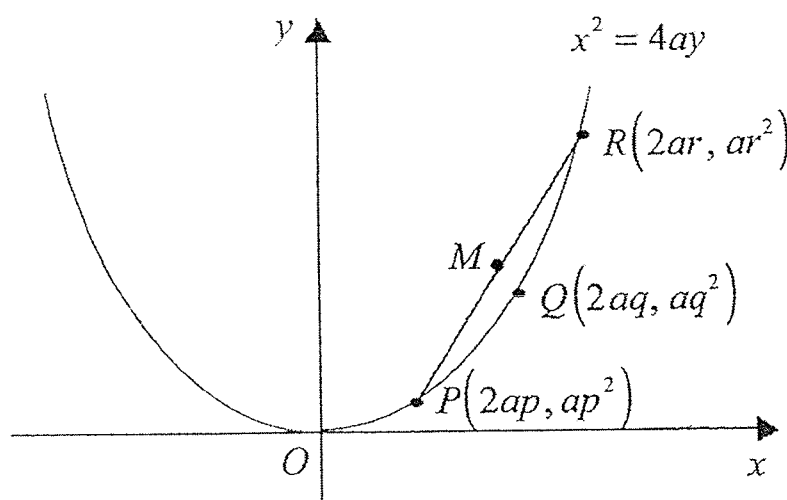
(b) Calculate the acute angle between the lines $x + y = 4$ and $3x - 4y = 2$

3

(c) Solve the inequality : $\frac{2}{x} \geq x - 1$

3

(d)



$Q(2aq, aq^2)$ is a fixed point on the parabola $x^2 = 4ay$ where $a > 0$. $P(2ap, ap^2)$ and $R(2ar, ar^2)$ are variable points which move on the parabola such that the chord PR is parallel to the tangent at Q .

(i) Show that $p + r = 2q$

3

(ii) Hence show that the coordinate of the midpoint, M is given by $(2aq, \frac{a(r^2 + p^2)}{2})$

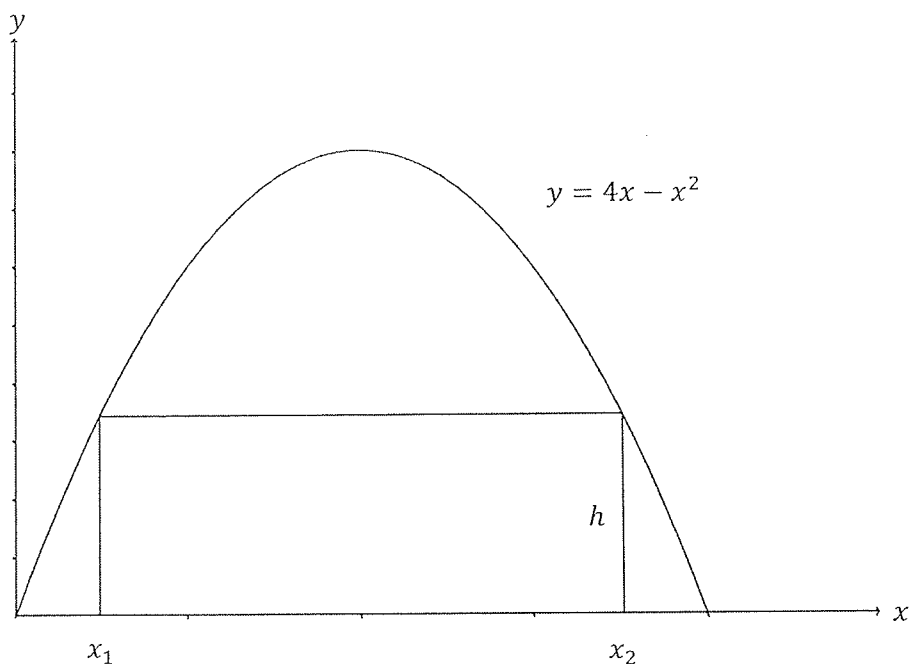
1

(iii) Explain why $r^2 + p^2 \geq 2q^2$

1

Question 10 (12 marks)

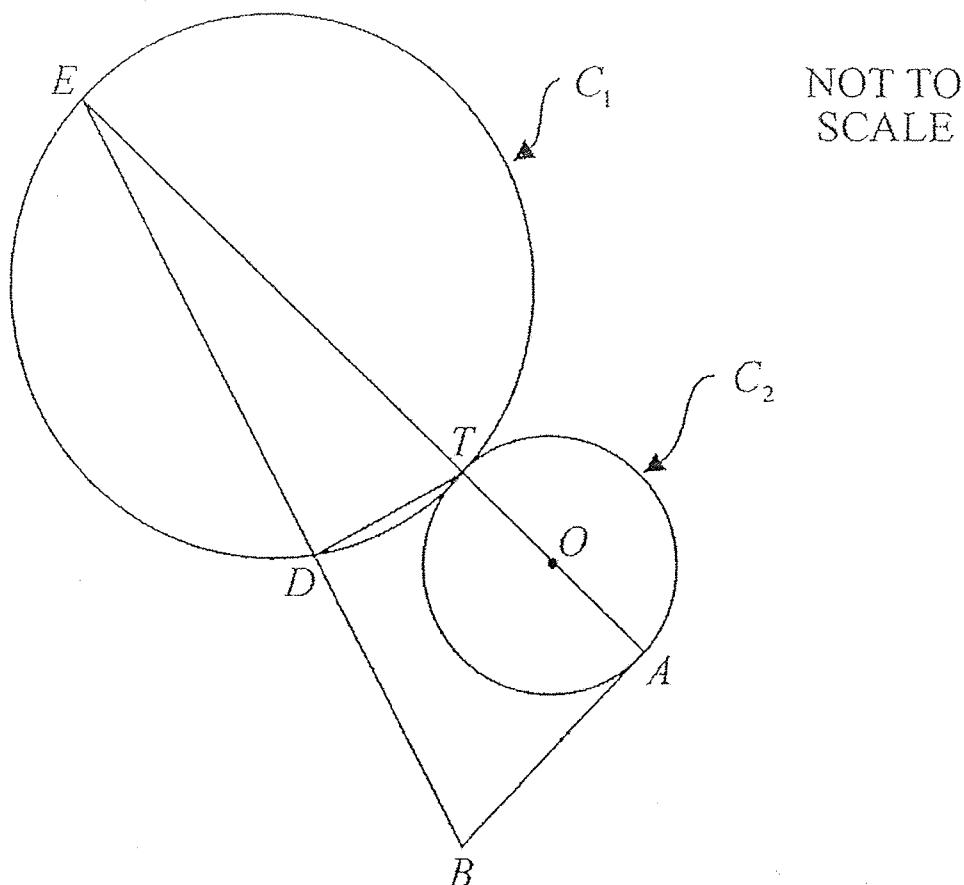
- (a) A rectangle has two of its vertices on the curve $y = 4x - x^2$ and the other two on the x axis in the interval $0 \leq x \leq 4$.



- (i) If the height of the rectangle is ' h ' units, show that its area is $2h\sqrt{4-h}$ units². 2
- (ii) Hence find the height of the rectangle when the area is a maximum. 3
- (b) Given the relation: $y = \pm \frac{x}{2}\sqrt{4-x^2}$
- and also given that $\frac{dy}{dx} = \pm \frac{2-x^2}{\sqrt{4-x^2}}$
- (i) Find the domain of this graph. 2
- (ii) Find the point(s) on the curve where the tangent(s) are horizontal 2
- (iii) Where on the curve are the tangents vertical (i.e. when the gradient is undefined)? 1
- (iv) Find the gradient(s) at the point $(0, 0)$. 1
- (v) Hence draw a *rough* sketch of the curve. 1

Question 11 (12 marks)

(a)



Two circles C_1 and C_2 touch at T .

The line AE passes through O , the centre of C_2 , and through T .

The point A lies on C_2 and E lies on C_1 .

The line AB is tangent to C_2 at A , D lies on C_1 and BE passes through D .

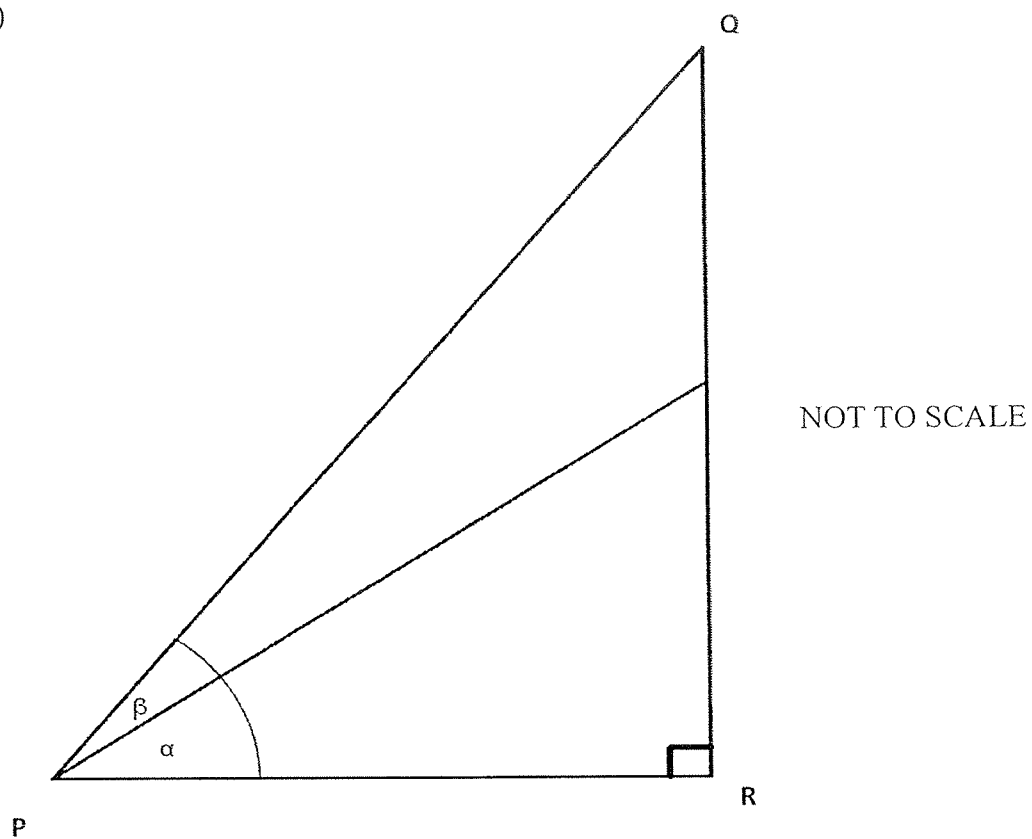
The radius of C_1 is R and the radius of C_2 is r .

(i) Explain why $\angle EDT = 90^\circ$ 2

(ii) If $DE = 2r$, show that $EB = \frac{2R(R+r)}{r}$ 2

Question 11 continued overleaf

(b)



In the diagram $\alpha > \beta > 0$, and $\frac{\cos(\alpha+\beta)}{\cos(\alpha-\beta)} = \frac{4}{5}$

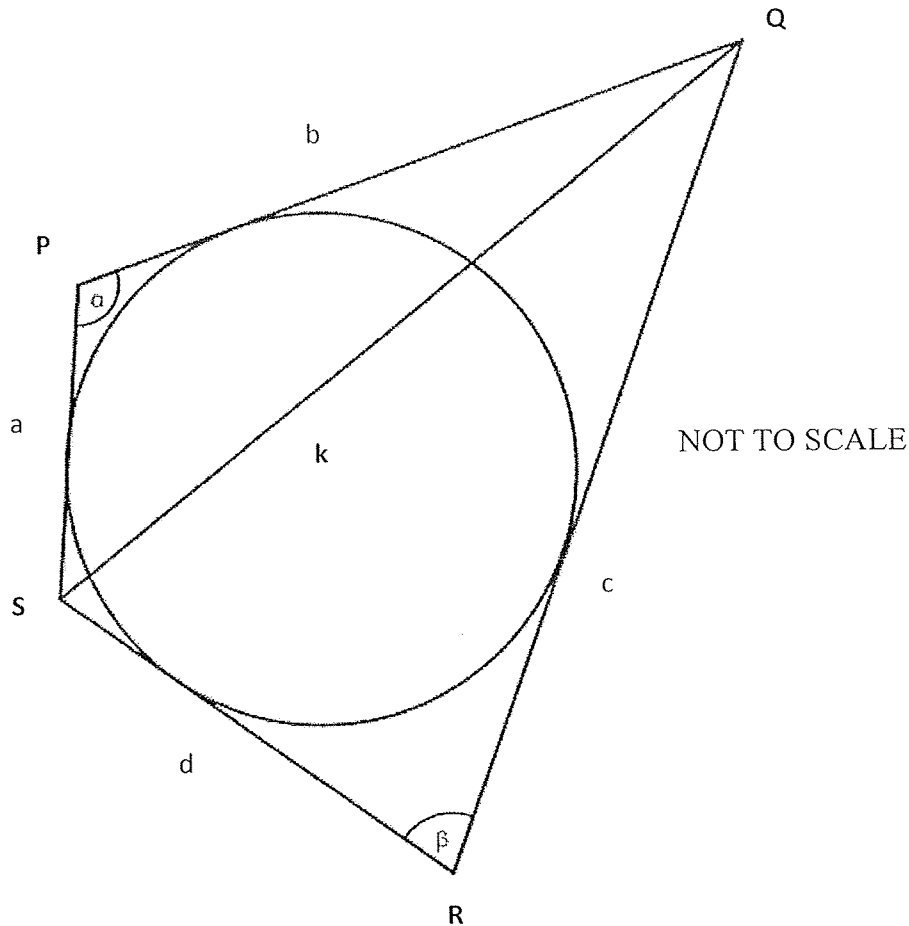
- (i) Show that $\tan \alpha \tan \beta = \frac{1}{9}$ 2
- (ii) If $PR = QR$, Show that $9(\tan \alpha + \tan \beta) = 8$ 3
- (iii) Hence, find the value of α and β to the nearest minute. 3

Question 12 (9 marks)

A convex quadrilateral $PQRS$ has side lengths a, b, c, d and an area of $\sqrt{abcd}$ units².

The quadrilateral has an inscribed circle as shown in the diagram below.

Let k be the length of the diagonal SQ and $\angle SPQ = \alpha$ and $\angle SRQ = \beta$.



- | | | |
|-------|--|---|
| (i) | By using the <i>cosine rule</i> , find two different relationships for k^2 . | 1 |
| (ii) | By using trigonometry find an expression for the area of the quadrilateral $PQRS$. | 1 |
| (iii) | Explain why $(a - b)^2 = (c - d)^2$ | 2 |
| (iv) | By using part (i) and (iii), show that
$ab(1 - \cos \alpha) = cd(1 - \cos \beta)$ | 2 |
| (v) | Hence, show that the quadrilateral $PQRS$ is a cyclic quadrilateral | 3 |

END OF EXAMINATION

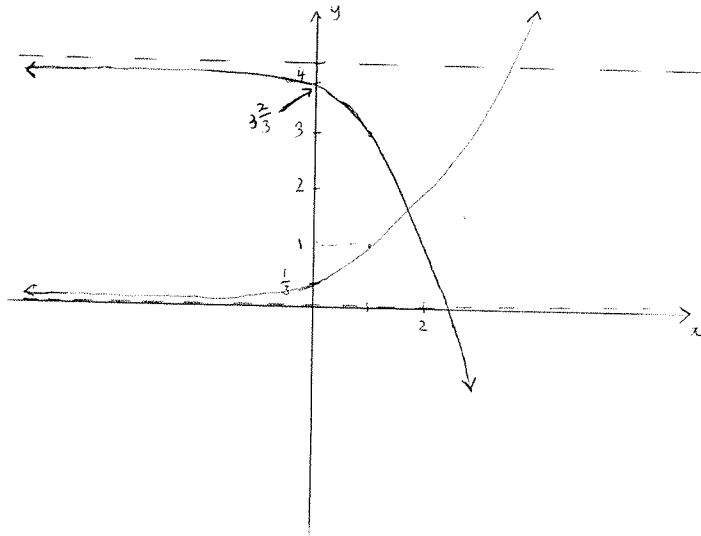
SECTION 1

1. D
2. B
3. B
4. D
5. C

SECTION II

Question 6 (8 marks)

a)

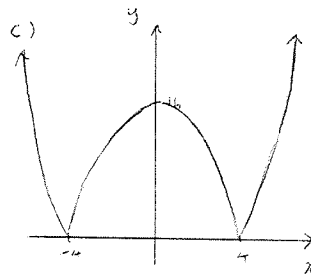


b) $f(x) = \frac{3x^2 - 2}{x^3}$, $f(-x) = \frac{3(-x)^2 - 2}{(-x)^3}$

$$= -\frac{3x^2 - 2}{x^3}$$

$$= -f(x)$$

$\therefore f(x)$ is an odd function



Question 8

a) when $x = -1$

(i) $(-1)^2 - k(-1) + 4 = 0$

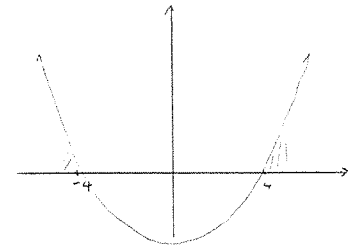
$$k + 5 = 0$$

$$k = -5$$

(ii) Real roots occur when $\Delta \geq 0$

$$\Delta = (-k)^2 - 4 \times 1 \times 4$$

$$= k^2 - 16 \geq 0$$



$\Delta \geq 0$ when

$$x \leq -4 \text{ or } x \geq 4$$

(iii) let one root be x

$\therefore$ 2nd root is $2x$

Product of roots: $2x^2 = 4$

$$x^2 = 2$$

$$x = \pm\sqrt{2}$$

From sum of roots: $x + 2x = k$

$$k = 3x$$

$$= \pm 3\sqrt{2}$$

b) $-12y = x^2 - 6x - 3$

$$= x^2 - 6x + 9 - 9 - 3$$

$$= (x - 3)^2 - 12$$

$$-12y + 12 = (x - 3)^2$$

$$-12(y - 1) = (x - 3)^2$$

(i) focal length = 3

vertex is at $(3, 1)$

parabola is concave down

$\therefore$ directrix is at $y = 4$

(ii) vertex $(3, 1)$

focal length = 3

focus: $(3, -2)$

Question 4

a) $|2x-3| = 3x+7$

This implies that $3x+7 \geq 0$

$$x \geq -\frac{7}{3}$$

Now we solve

$$2x-3 = 3x+7 \quad \& \quad 2x-3 = -3x-7$$

$$x = -10 \quad \text{OR} \quad 5x = -4$$

$$x = -\frac{4}{5}$$

Since $x \geq -\frac{7}{3}$

then only solution is

$$x = -\frac{4}{5}$$

b) Let $l_1: x+y=4$

$$l_2: 3x-4y=2$$

$$\therefore m_1 = -1 \quad \text{and} \quad m_2 = \frac{3}{4}$$

acute angle between two lines

can be found by:

$$\tan \theta = \left| \frac{-1 - \frac{3}{4}}{1 + (-\frac{3}{4})} \right|$$

$$= \left| \frac{-\frac{7}{4}}{\frac{1}{4}} \right|$$

$$= 7$$

$$\therefore \theta = \tan^{-1} 7$$

$$\approx 81^\circ \quad (\text{to nearest degree})$$

c) $\frac{2}{x} \geq x-1$

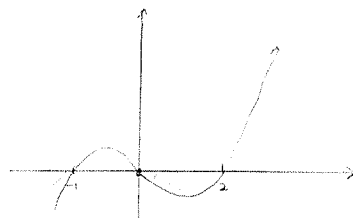
multiplying by the square of denominator to both sides

$$2x \geq x^3 - x^2 \quad \text{where } x \neq 0$$

$$x^3 - x^2 - 2x \leq 0$$

$$x(x^2 - x - 2) \leq 0$$

$$x(x-2)(x+1) \leq 0$$



$$x \leq -1 \quad \text{OR} \quad 0 < x \leq 2$$

as $x \neq 0$

d) Gradient at Q: $\frac{dy}{dx} \times \frac{dq}{dx}$

$$= 2aq \times \frac{1}{2a}$$

$$= q$$

Now chord PQ has

$$\text{gradient at: } \frac{ar^2 - ap^2}{2ar - 2ap}$$

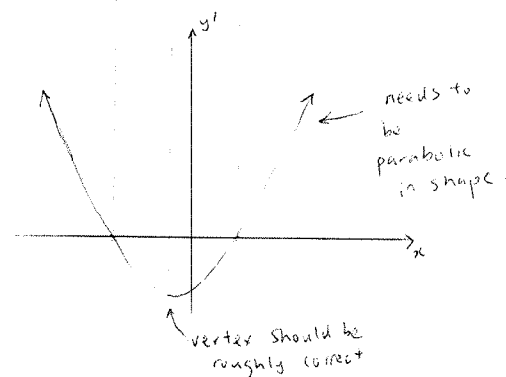
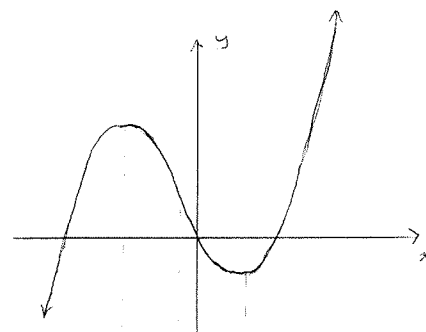
$$= \frac{a(r+p)(r-p)}{2a(r-p)}$$

$$= \frac{r+p}{2}$$

Since they have equal gradients

$$\frac{r+p}{2} = q \Rightarrow p+r=2q$$

Question 7



b) $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$f(x+h) = 3(x+h)^2 - 2(x+h)$$

$$= 3(x^2 + 2xh + h^2) - 2x - 2h$$

$$= 3x^2 + 6xh + 3h^2 - 2x - 2h$$

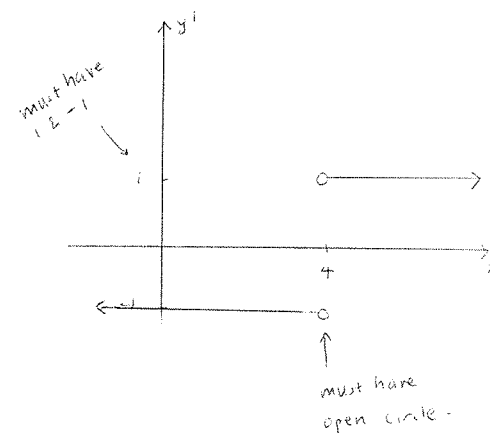
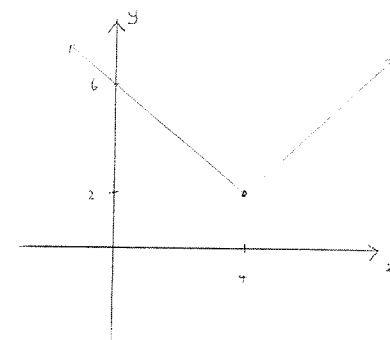
$$\therefore \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{(3x^2 + 6xh + 3h^2 - 2x - 2h) - (3x^2 - 2x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 - 2h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(6x + 3h - 2)}{h}$$

$$= \lim_{h \rightarrow 0} 6x + 3h - 2$$

$$= 6x - 2$$



$$\begin{aligned} \text{c) i) } y &= 3x^{\frac{1}{2}} \\ &= \frac{3}{2} x^{-\frac{1}{2}} \\ &= \frac{3}{2\sqrt{x}} \end{aligned}$$

$$\text{ii) } y = (4-3x)^5 (2x+1)$$

$$\begin{aligned} \frac{dy}{dx} &= (4-3x)^5 \cdot \frac{d}{dx}(2x+1) + (2x+1) \cdot \frac{d}{dx}(4-3x)^5 \\ &= 2(4-3x)^5 + (2x+1) \cdot 5 \cdot (-3) \cdot (4-3x)^4 \\ &= 2(4-3x)^5 - 15(4-3x)^4(2x+1) \\ &= (4-3x)^4 [2(4-3x) - 15(2x+1)] \\ &= (4-3x)^4 [8-6x-30x-15] \\ &= (4-3x)^4 [-36x-7] \end{aligned}$$

$$\text{iii) } y = \frac{x^2+2}{x+4}$$

$$u = x^2+2 \quad v = x+4$$

$$\frac{du}{dx} = 2x \quad \frac{dv}{dx} = 1$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{(x+4) \cdot 2x - (x^2+2)}{(x+4)^2} \\ &= \frac{2x^2+8x-x^2-2}{(x+4)^2} \\ &= \frac{x^2+8x-2}{(x+4)^2} \end{aligned}$$

Question 9 (cont)

$$\text{ii) Midpoint: } M \left(\frac{2ar+2ap}{2}, \frac{ar^2+ap^2}{2} \right)$$

$$M \left(a(r+p), \frac{a(r^2+p^2)}{2} \right)$$

$$\text{Since } r+p=2q \text{ (from (i))}$$

$$M \left(2aq, \frac{a(r^2+p^2)}{2} \right) \quad [1]$$

iii) Since the x value of M is equal to the x value of Q .

M lies directly above Q . Since P and R lie to the left and right of Q , M is always above the point Q . Although equality exists when P, Q, R are the same points.

[1] for reasonable explanation.

$$\therefore \frac{a(r^2+p^2)}{2} \geq aq^2$$

$$r^2+p^2 \geq 2q^2 \quad (\text{as } a > 0)$$

Question 10

i) To find x_1 and x_2 . Find the intersection of the parabola

$$y = -x^2 + 4x \text{ with } y = h$$

hence x_1 and x_2 are the roots of $x^2 - 4x + h = 0$

$$\text{where } x_1 + x_2 = 4 \text{ and } x_1 x_2 = h$$

The length of the rectangle can be expressed as $x_2 - x_1$.

$$\begin{aligned} \text{We know } (x_2 - x_1)^2 &= (x_1 + x_2)^2 - 4x_1 x_2 \\ &= 16 - 4h \\ &= 4(4 - h) \end{aligned}$$

$$\therefore x_2 - x_1 = 2\sqrt{4 - h}$$

$$\begin{aligned} \text{Area of rectangle (A)} &= h(x_2 - x_1) \\ &= 2h\sqrt{4 - h} \end{aligned}$$

Note: Quadratic formula to find x_1 and x_2 can be used.

$$\begin{aligned} \text{ii) } \frac{dA}{dh} &= 2h \times \frac{d}{dh}(4 - h)^{\frac{1}{2}} + (4 - h)^{\frac{1}{2}} \times \frac{d}{dh}(2h) \\ &= -\frac{h}{\sqrt{4 - h}} + 2\sqrt{4 - h} \\ &= \frac{2(4 - h) - h}{\sqrt{4 - h}} \\ &= \frac{8 - 3h}{\sqrt{4 - h}} \end{aligned}$$

Stationary point occurs when $\frac{dA}{dh} = 0$

$$\text{i.e. } 8 - 3h = 0$$

$$3h = 8$$

$$h = \frac{8}{3}$$

Testing the nature of the stationary point

h	2	$\frac{8}{3}$	3
$\frac{dA}{dh}$	$\frac{2}{\sqrt{2}}$	0	-1

For testing nature students must give actual output! They can use first or 2nd derivative, but check for algebraic mistakes for 2nd derivative.

$\therefore$ maximum stationary point when $h = \frac{8}{3}$

$h = 8$ is the height of rectangle when the area is a maximum.

$$\text{b) i) Domain: } 4 - x^2 \geq 0$$

$$-2 \leq x \leq 2$$

ii) tangents horizontal when $\frac{dy}{dx} = 0$

$$\text{i.e. } 2 - x^2 = 0$$

$$x^2 = 2$$

$$x = \pm\sqrt{2}$$

$\therefore$ the points are

$$(\sqrt{2}, 1), (\sqrt{2}, -1), (-\sqrt{2}, 1), (-\sqrt{2}, -1)$$

iii) when gradient is undefined:

$$4 - x^2 = 0$$

$$\text{i.e. when } x = \pm 2$$

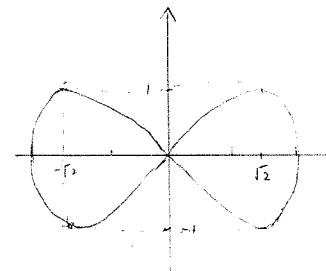
$$(2, 0), (-2, 0)$$

iv) when $x = 0$

$$\frac{dy}{dx} = \pm 1$$

$$\therefore \text{gradient} = m = \pm 1$$

v)



Question 11

(i) When two circles touch, the two centres and the point of contact are collinear, hence AE must pass through the centre of C_1 .

$\therefore \angle EDT = 90^\circ$ (angle in a semicircle)

(ii) $\because \angle OAB = 90^\circ$ (tangent AB meet radius OA at right angles)

and $\angle EDT$ is common

$\triangle EDT \sim \triangle EAB$ (equiangular)

$$\frac{x}{2R} = \frac{2R+2r}{2r} \quad (\text{corresponding sides in proportion})$$

$$x = \frac{2R \cdot 2(R+r)}{2r}$$

$$= \frac{2R(R+r)}{r}$$

(i) $\frac{\cos(x+\beta)}{\cos(x-\beta)} = \frac{4}{5}$

$$5 \cos(x+\beta) = 4 \cos(x-\beta)$$

$$5 \cos x \cos \beta - 5 \sin x \sin \beta = 4 \cos x \cos \beta + 4 \sin x \sin \beta$$

$$\cos x \cos \beta = 9 \sin x \sin \beta$$

$$\text{divide through by } 9 \cos x \cos \beta$$

$$\tan x \tan \beta = \frac{1}{9}$$

(ii) If $PR = QR$ then $\tan(x+\beta) = 1$

$$\frac{\tan x + \tan \beta}{1 - \tan x \tan \beta} = 1$$

from part (i)

$$\tan x + \tan \beta = 1 - \tan x \tan \beta$$

$$= 1 - \frac{1}{9}$$

$$= \frac{8}{9}$$

$$\therefore 9(\tan x + \tan \beta) = 8$$

(iii) From part (i) and (ii)

we can let $\tan x$ and

$\tan \beta$ be the roots of

$$x^2 - \frac{8}{9}x + \frac{1}{9} = 0$$

$$\therefore 9x^2 - 8x + 1 = 0$$

$$x = \frac{8 \pm \sqrt{64-36}}{18}$$

$$\tan x = \frac{8+2\sqrt{7}}{18}$$

$$x = 36^\circ 27'$$

$$\tan \beta = \frac{8-2\sqrt{7}}{18}$$

$$\beta = 8^\circ 53'$$

Question 12

i) $k^2 = a^2 + b^2 - 2ab \cos \alpha$

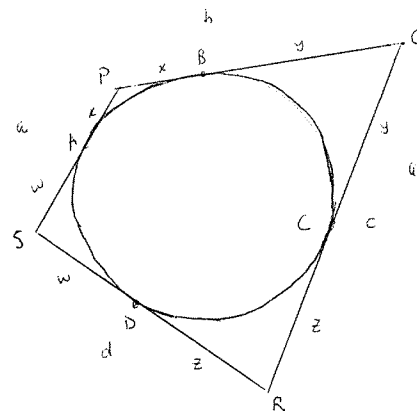
$$k^2 = c^2 + d^2 - 2cd \cos \beta$$

ii) Area PQRS = Area $\triangle PQS$ + Area $\triangle SRQ$

$$= \frac{1}{2} ab \sin \alpha + \frac{1}{2} cd \sin \beta$$

$$= \frac{1}{2} (ab \sin \alpha + cd \sin \beta) = \sqrt{abcd}$$

iii)



Let the point of contact be

A, B, C, D

and $AP = PB = x$ (Two tangents from an external point are equal in length)

Similarly $BQ = QC = y$

$$RC = RD = z$$

$$SD = SA = w$$

$$(a-b)^2 = [(w+x) - (x+y)]^2$$

$$= [w-y]^2$$

$$= [-(y-w)]^2$$

$$= (y-w)^2$$

$$= (y+z) - (w+z)]^2$$

$$= (c-d)^2$$

iv) From (i)

$$a^2 + b^2 - 2ab \cos \alpha = c^2 + d^2 - 2cd \cos \beta$$

$$\text{subtract } (a-b)^2 = (c-d)^2$$

$$2ab - 2ab \cos \alpha = 2cd - 2cd \cos \beta$$

$$ab(1 - \cos \alpha) = cd(1 - \cos \beta)$$

v) From (ii)

$$\frac{1}{2} (ab \sin \alpha + cd \sin \beta) = \sqrt{abcd}$$

Squaring both sides we get

$$a^2 b^2 \sin^2 \alpha + c^2 d^2 \sin^2 \beta + 2abcd \sin \alpha \sin \beta = 4abcd$$

$$a^2 b^2 (1 - \cos^2 \alpha) + c^2 d^2 (1 - \cos^2 \beta) + 2abcd \sin \alpha \sin \beta = 4abcd$$

$$ab \cdot ab(1 - \cos^2 \alpha) + cd \cdot cd(1 - \cos^2 \beta) + 2abcd \sin \alpha \sin \beta = 4abcd$$

From (iii)

$$ab(1 + \cos \alpha)cd(1 - \cos \beta) + cd \cdot ab(1 - \cos \alpha)cd(1 + \cos \beta) + 2abcd \sin \alpha \sin \beta = 4abcd$$

dividing through by $abcd$

$$(1 + \cos \alpha)(1 - \cos \beta) + (1 - \cos \alpha)(1 + \cos \beta) + 2 \sin \alpha \sin \beta = 4$$

$$1 - \cancel{\cos \beta} + \cancel{\cos \alpha} - \cos \alpha \cos \beta + 1 + \cancel{\cos \beta} - \cancel{\cos \alpha} - \cos \alpha \cos \beta + 2 \sin \alpha \sin \beta = 4$$

$$2 - 2 \cos \alpha \cos \beta + 2 \sin \alpha \sin \beta = 4$$

$$2 - 2 \cos(\alpha + \beta) = 4$$

$$\cos(\alpha + \beta) = -1$$

$$\therefore \alpha + \beta = 180^\circ$$

$\therefore$ quadrilateral is cyclic as opposite angles
are supplementary